

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes.

Does R2 show up as an OSPF neighbor on R3? No.

What does this information tell you?

This means that R2 only establishes an OSPF neighbor relationship with R1 and exchanges route updates, but does not establish a neighbor relationship with R3. Therefore, R3 must learn and route to R2 network through R1.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The cost is 15. The sum of the cost of the Serial interface (64) of the R3 sender and the cost of the GigabitEthernet interface (1) of the R2 destination is obtained.

Is R2 listed as an OSPF neighbor to R3? Yes.

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default reference bandwidth is 100 Mb/s, which assigns a cost of 1 to all links operating at 100 Mb/s or faster. Changing it to a higher value allows OSPF to accurately differentiate and calculate costs for modern, higher-speed links. _____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

To 192.168.3.0/24, the cost is 782. R1's S0/0/1 bandwidth was set to 128 (Cost 781), plus R3's G0/0 (Cost 1). ($781 + 1 = 782$).

To 192.168.23.0/30, the cost is 845. R1's S0/0/1 (Cost 781) plus R3's unmodified S0/0/1 (Cost 64). ($781 + 64 = 845$). _____

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new cost is 1562. Because the bandwidth 128 command was applied to all serial interfaces, each serial link now has a cost of 781. The route travels over two serial links (R1's outgoing interface + R2/R3's outgoing interface): $781 + 781 = 1562$. _____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The direct path via S0/0/1 was manually assigned a cost of 1565, making its total cost $1565 + 1 = 1566$. The indirect path via R2 costs 1563 ($781 + 781 + 1$). OSPF routes traffic through R2 because 1563 is a lower cost than 1566. _____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

__This action provides for predictability and consistency throughout the overall OSPF routing domain. When the router ID is associated only with a physical interface, it creates an opportunity for the router ID to change whenever that link fails; therefore, a static IP or loopback interface should be used to ensure stability.____

2. Why is the DR/BDR election process not a concern in this lab?

__The serial links connecting the routers are Point-to-Point networks. OSPF does not require or conduct DR/BDR elections on Point-to-Point links.____

3. Why would you want to set an OSPF interface to passive?

__It prevents OSPF routing updates (Hello packets) from being sent out of the specified interface. This reduces unnecessary traffic and CPU load on LANs that do not contain other OSPF routers.____

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

__The use of OSPF Areas.____

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

__network 10.0.0.0 0.255.255.255 area 0____

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

__EIGRP. Routers use Administrative Distance (AD) to decide between different routing protocols. EIGRP has the lowest default AD (90) compared to OSPF (110) and RIP (120), so its route is preferred regardless of the metric.____